

Unit 1 Mini-Cases

Principles of Capital Markets

Winter Term 2025/26

Please upload your solutions and indicate which Mini-Cases (MCs) you have solved before **October 8, 2025 at 6am**. To do so, go to the assignment section on Canvas and choose “Unit 1 Indication and Submission of MCs”. Then answer the true/false questions (Questions 1 to 6) and upload your solutions file (Question 7). **There will be no deadline extensions! Only complete submissions containing answers to the T/F questions and a solutions upload will be considered, your last complete attempt counts!** You can either write your solutions using software or scan your handwritten solutions. Please note that you may only submit one solutions file. If your solutions consists of more than one file (e.g. 1 pdf and 1 Excel file), please upload a .zip file. Please make sure to hand in only one file of each file type (you might have to merge your scans to one pdf file). Uploaded files will be checked for plagiarism. The solutions in your file should be self-explanatory - not only for the instructor, but for other students as well. Indicate only Mini-Cases you feel confident to present.

Important remark for the presentations: Make sure to provide a concise explanation of your solutions. In particular, when applying any formula from the lecture materials or other sources, you may be asked to explain the structure of your solutions and provide the intuition and underlying concepts behind the formula used.

Mini-Case 1.1 (pdf)

You are thinking about insuring your trailer over the next 5 years. You have the following choices:

1. pay beginning-of-year premia worth EUR 500 each, i.e. the first premium is paid today and there are going to be 4 more payments
2. prepay a lump sum of EUR 2,500 for the entire 5 years of coverage or
3. pay a sum of EUR 400 today, a sum of EUR 1,500 in two years and EUR 600 in four years.

Answer the following questions:

- (a) Assuming a constant interest rate (R) of 3.5% per year compounded annually, what is the most favorable choice?
- (b) Assume that R decreases. How does this affect the present value of each option?

- (c) You decide to neglect the options above but pay for any possible losses or damages out of your own pocket. For this purpose, you put EUR 2,350 in a savings account today paying a constant interest rate of 3.5% per year compounded annually for the next 5 years. How much will your savings be after 5 years if there are no events of damage or loss?

Mini-Case 1.2 (pdf)

Consider an investor with utility function $U(W) = \sqrt{W}$. Perform the following tasks:

- Calculate the utility of wealth of the investor for wealth levels 1, 2, and 4.
- Use your results from (a) to plot the utility function (wealth on the x-axis and utility of wealth on the y-axis).
- Calculate the marginal utility for wealth levels 1, 2, and 4. Is marginal utility increasing or decreasing?
- Describe the risk preference of the investor.
- Find two examples for a utility function representing a different risk preference than the one above.

Mini-Case 1.3 (pdf)

The following table includes information on two distinct lotteries X and Y that you can choose to participate in. It includes information on the number of possible “states” (scenarios), the probability for each state and the wealth you get in each state. You can only participate in one of the two lotteries and there is no fee to be paid to participate in either of them. Your preferences are best described by the utility function $U(W) = W^2$.

lottery	state (number)	probability	wealth
X	1	0.5	50
	2	0.3	75
	3	0.2	150
Y	1	0.6	25
	2	0.4	150

- Calculate the expected wealth of lottery X and Y .
- Calculate the expected utility of lottery X and Y .
- Do you participate in lottery X or lottery Y when your goal is to
 - maximize your wealth?
 - maximize your utility?

Explain the result and your decision-making process.

Mini-Case 1.4 (Excel or R or pdf)

You are the owner of a family farm with a wealth of EUR 217,500. You must choose between sitting this harvesting season out and investing your money in a safe money market fund paying 3.5% p.a. compounded annually or use the money to plant corn. If there is rain, planting will result in EUR 460,000 in wealth in one year. If there is a drought, planting will give EUR 50,000 in wealth. The probability of rain and drought is 55% and 45%, respectively. You are risk averse, and your preference for family wealth (W) is specified by the relationship $U(W) = \sqrt{W}$.

- (a) Which option do you choose if you want to maximize expected wealth?
- (b) Which option do you choose if you want to maximize expected utility?
- (c) As a third choice, you can plant a mix of standard and drought-resistant corn, i.e. 50% of each corn type. Drought resistant corn is more expensive and will result in lower wealth of EUR 290,000 in case of rain compared to standard corn, but EUR 140,000 in wealth in case of a drought. Repeat your calculations for (a) and (b) above considering the plant mix and check for each case if your decision changes.

Mini-Case 1.5 (Excel or R)

Your generous parents have offered to support you financially during your studies in London (UK). You will receive the following payments: GBP 15,000 today, GBP 15,500 one year from today, GBP 16,000 in two years, GBP 16,500 in three years, GBP 17,000 in four years and GBP 17,500 in five years.

- (a) Calculate today's value of the financial support in GBP using the current UK interest rate term structure. You can find the current UK interest rates under <https://markets.ft.com/data/bonds> ($\rightarrow$ "Table").

Hint: The interest rate term structure (yield curve) provides the different interest rates (usually expressed per annum) for different maturities. You can assume that the given rates are spot rates.

- (b) Plot and interpret the current UK interest rate term structure using all the information under the above link.

Mini-Case 1.6 (Excel or R)

Download monthly closing prices for the stock market indices ATX, DAX, EUROSTOXX 50, Nasdaq Composite and HangSeng provided in the assignment section of unit 1 ($\rightarrow$ "MarketDataUnit1.csv")¹. Use the entire observation period for your analyses.

- (a) Summarize the market coverage of the indices provided.
- (b) Plot the historic prices for each index in a separate diagram.

¹Data Source: LSEG

- (c) Compare and analyze the market developments for the five indices.
- (d) Calculate monthly returns for each index.

Hint: The monthly returns can simply be calculated as $\frac{P_t - P_{t-1}}{P_{t-1}}$, where P_t denotes the adjusted closing price for month t .

- (e) Plot the monthly returns for each index using a histogram. Based on the five histograms, compare and characterize the distributions of the returns for the five indices.